

Research Advanced in Human Pose Estimation based on Deep Learning

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Abstract. Human pose estimation constitutes a crucial task within the realm of computer vision. Its objective is to precisely identify and locate the main features of the human body as shown in images or videos, thereby enabling the understanding and analysis of human behavior. In the early stages, pose estimation efforts predominantly focused on single-person scenes, relying mainly on template matching, direct regression, and detection methods. Thanks to the rapid progress of deep learning technologies, research on human pose estimation has evolved from single-person scenarios to complex multi-person ones. Multi-person pose estimation can be separated into top-down and bottom-up approaches. The top-down method initially detects the human body and then conducts pose estimation. Although relatively straightforward to implement, it has limitations when dealing with multi-person scenes and occlusion situations. In contrast, the bottom-up method first detects main features and then assigns them to different individuals, proving more effective in handling occlusion and multi-person scenes. Considering the above factors, this paper undertakes a comprehensive and in-depth study on human pose estimation based on deep learning, covering multiple aspects such as single-person pose estimation, multi-person pose estimation, datasets, evaluation metrics, algorithms, and experimental methods. Additionally, it discusses the existing problems and future development directions from perspectives including improving the model architecture, optimizing data processing, and integrating multi-modal information. Moreover, recent advancements in deep learning architectures, including convolutional neural networks and recurrent neural networks, have further enhanced the performance and accuracy of human pose estimation. The combination of different modalities, such as depth information and infrared imaging, also holds promise for addressing some of the challenges in complex scenes.

Keywords: Human Pose Estimation, Top-down Approach, Bottom-up Approach, Single-stage Approach, Transformer-based Approach.

1. Introduction

Human posture estimation is a computer vision technique, which aims to identify and localize specific body landmarks such as head, wrists, shoulders, knees, etc. It helps people understand and analyze the human behavior of pedestrians. Driven by deep learning, this technology utilizes powerful feature extraction mechanisms of deep neural networks to enhance the precision of recognizing human postures in pictures. Human posture estimation can support numerous tasks in the realm of computer vision, including action recognition [1-2], augmented reality, virtual reality [3], health monitoring, and human-computer interaction. Given its cost-effectiveness and flexibility, it is considered as a fundamental task in computer vision and attracts increasing research attention.

Existing posture estimation encompasses single-person and multi-party categories, based on the number in an image. Single person-posture estimation handles scenarios with a single pedestrian and only requires predicting the body landmark positions. Early single-person pose estimation methods are mainly based on template matching, such as using the pictorial structure and spring model to predict key points. With the progress of deep learning technology, single person pose estimation using

deep learning models began to be performed by researchers. Single person pose estimation includes techniques based on both detection and regression, employing landmark point coordinate calculation algorithms such as DeepPose and Softmax Regression. Instead of directly regressing the key point coordinates, the detection-based method first finds the regions of the image where the critical point may exist and afterwards decodes the coordinates of critical points by detecting these regions. The representative work is using heatmap prediction methods such as Convolutional Pose Machine (CPM).

For the multi-person posture estimation, beyond the single-person posture estimation task, it also requires matching body landmarks with different pedestrians and resolving issues like difficulty distinguishing pedestrians' mutual occlusion [4]. Multi-person posture estimation is mainly grouped into the following categories:

(1) Top-down approaches. In this approach, all people in the picture are detected at first, and then this approach does a pose estimation for everyone. Though relatively simple to implement, its performance is limited when dealing with multi-person scenes and occlusions with significant increase of computational complexity and inference time. Researchers are exploring how to advance the precision of human detection and how to deal with the human inside the detection frame more efficiently, especially when there are overlaps and occlusions between the human bodies. (2) Bottom-up approaches. Bottom-up methods find key points of bodies of all people in the picture at first and then the key points are assigned to different human bodies through algorithms such as clustering. These methods are more effective for dealing with occlusions and multi-person scenes without additional detection and posture estimation operation, which helps improve reasoning speed. (3) Single-stage approaches. The single-stage methods directly predict the human pose from the input images without going through multi-stage processing, which usually holds the advantages of high efficiency and real-time performance. (4) Transformer-based approaches. Transformer [5] is a kind of deep learning model and is used for both single-person and multi-person pose estimation. The kernel of Transformer is the Self-Attention mechanism, and this mechanism allows the model to consider each position of the sequence and all the other positions, thus capturing dependencies within the sequence. In terms of computational efficiency, despite the Self-Attention mechanism having brought challenges to the amount of calculation, the Transformer design allows efficient parallel processing to optimize the computation efficiency of the models.

Focusing on the above four aspects, this paper provides a detailed review of advanced multi-person pose estimation methods which base on deep learning. Specifically, Section 2 introduces representative approaches, including their research problems, design ideas, and key processes. Subsequently, this paper also introduces commonly used pose estimation datasets and evaluation indicators in Section 3, and quantitatively compares the performance differences of different categories of methods. In Section 4, the problems that still exist in the field of human pose estimation are discussed and its future development direction is expected. Finally, Section 5 sums up the primary work and contributions of this paper.

2. Method

2.1. Top-down Approach

The top-down approach usually refers to detect all the people in the picture and then do posture estimation for each individual. Since detection performance plays a crucial role, researchers are exploring how to improve the accuracy of human body detection and deal with the human inside the detection frame more efficiently, especially when there are overlaps and occlusions between the human bodies. CNN (Region-based Convolutional Network) combines AlexNet with an alternative search tactic to obtain the detection, while its speed is limited due to the redundant feature extraction for each proposal. To solve this issue, Fast R-CNN extracts the feature of whole image and then crop the feature of each proposal, while Faster R-CNN designs the regional proposed network (RPN) to replace the selective search. Mask R-CNN proposes to extract features with the feature pyramid network and replace the RoIPool layer with the RoIAlign layer, further enhancing the precision and

efficiency of Faster-RCNN. In addition, the Regional Multi-Person Pose Estimation (RMPE) framework is proposed to resolve the problem of imprecise human bounding box detection. At the same time, unbiased data processing (UDP) helps analyze the common biased attitude estimation of the human pose estimation according to the unit length rather than pixel processing data. The Graph-RCNN model is a two-stage framework that uses a heatmap regressor to locate key points and designs a customized human pose refinement module to improve accuracy. Graph-RCNN introduces a residual step network (RSN) to seek a better balance between local and global feature expression. At the same time, for the purpose of alleviate the problem of insufficient key point detection in crowded multi-person pose scenes, Graph-RCNN proposes a top-down distribution of key point coordinates, which solves the quantization deviation by decoding the predicted key point heatmap into the two-dimensional coordinates (x , y) of the joints.

2.2. Bottom-up Approach

Bottom-up multi-person pose estimation methods first accurately identify the body parts in the image. Then, it can construct the overall skeleton structure of each person by deeply analyzing the spatial relationship between these parts. PAF (Part Affinity Fields) [6] proposes the affinity field to represent the relationship between key points, which can maintain high accuracy even for occluded and crowded persons. The affinity field describes the spatial relation and connection strength between one key point and another one. Key points detection, affinity field generation, graph searching using affinity field information, individual skeleton construction, and post-processing are the main process of PAF. The computational efficiency of this algorithm still needs optimization. OpenPose [7] directly delineates all people's key points in the imagery and bypassing reliance on a human detector based on PAF. Specifically, A deep neural network is used to predict partial affinity fields (PAFs), representing spatial connections between various body parts by OpenPose. In addition, the network uses end-to-end training, which enables direct prediction of key point positions and PAFs from image pixels. This algorithm achieves accurate multi-person pose estimation in real time with the help of innovative network structure and training methods.

MutiPoseNet [8] proposes a residual pose network, which uses feature pyramids to extract key point information and perform human detection. Then all the key points are fed into the residual pose network for clustering. MutiPoseNet significantly improves accuracy and speed, which make it potentially applicable in action recognition, human-computer interaction, and video surveillance analysis. But it still faces challenges when dealing with highly crowded scenes or extreme occlusion. PifPaf [9] extracts image features and generates probability maps of key points based on a deep neural network. It predicts connection relationships and determines optimal connectivity through partial affinity maps. PifPaf also introduces two head networks for PIF prediction (joint confidence, precise location, size) and PAF prediction (parts association), respectively. PifPaf can identify individual body parts and their connections with high accuracy and robustness, which has been applied in various tasks that need computational efficiency and real-time performance optimizations, such as action recognition, human-computer interaction, and video surveillance.

Higher HRNet [10] designs an improved feature pyramid network for multiple resolution monitoring, thus enhancing the effect of pose estimation on small-scale humans. The network structure is divided into four steps: using a standard convolutional neural network as the backbone network, extracting initial features from the input image; after feature extraction, multiple high-resolution branches are used to process feature maps; the feature pyramid network structure is employed to fuse different levels of feature maps; and the position of each key point is predicted. Higher HRNet has high computational resource requirements, making real-time implementation challenging on resource-limited devices.

2.3. Single-stage Approach

With the development of deep learning, single-stage methods have made significant progress in 2D pose estimation in recent years, which predicts the human pose from the input image without

going through multi-stage processing. This method usually has the advantages of high efficiency and real-time performance, and can quickly give the pose estimation results [11].

SPM (Hierarchical Structured Pose Representation) is an important technique in single-stage methods [12], which represents the human pose by constructing a hierarchical structure and can better capture the spatial relationship and structural information of the pose. In SPM, the human pose is represented as a hierarchical structure composed of key points, and each key point has a certain correlation with other key points. This hierarchical structure can help the model better understand the semantic information of the human pose, thereby improving the accuracy of pose estimation. A key advantage of hierarchical structured pose representation is that it can effectively integrate information at different scales. In 2D pose estimation [13], different parts have different scales and importance. For example, the head and hands are usually relatively small, while the body and legs are relatively large. SPM can adapt to these different scales of information through the hierarchical structure, thereby better capturing the details of the human pose. In addition, SPM can also use the spatial relationship between key points for pose estimation. By learning the relative positions and directions between key points, the model can better infer the pose of the human body, which helps improve the accuracy and robustness of pose estimation [13] when facing complex poses and occlusion situations. In practical applications, single-stage methods usually combine deep learning techniques, such as convolutional neural networks (CNN) and Transformer architectures. CNN is good at extracting local features of the image, while the Transformer architecture can capture the global context information. By combining the two, the single-stage method can make full use of the features of the image to improve the performance of pose estimation.

There are many other single-stage methods that have achieved good results in 2D pose estimation. For example, Simple Baseline [14] is a simple and easy-to-use model that predicts the positions of key points through a simple convolutional neural network. Hourglass combines information at different scales for prediction by stacking multiple hourglass-shaped modules to capture the details of the human pose. In general, single-stage methods have great potential in 2D pose estimation [14]. They can quickly give pose estimation results, and the accuracy and robustness have also been continuously improved. With the continuous development of deep learning technology, it is believed that single-stage methods will play an important role in more application scenarios in the future. However, single-stage methods still face some challenges. For example, in the face of complex backgrounds, occlusion, and variable poses, the accuracy of the model may be affected. In addition, how to effectively use limited training data to improve the performance of the model is also a problem that needs to be solved.

2.4. Transformer-based Approach

Transformer gradually becomes an important component for 2D human pose estimation due to its self-attention and global capacity, especially for the long-distance spatial relationship among key points. As one of the most representative methods, PRTR (Pose Recognition with Cascade Transformers) [15] leverages the powerful capabilities of the Transformer architecture, the pose estimation is gradually refined in a cascade manner, resulting in more accurate and robust pose recognition. Specifically, PRTR extracts a preliminary feature representation from the input image and then refines these features using the standard Transformer visual feature encoder. Based on the hierarchical feature maps, PETR regards pose estimation as an assembling-regression question and proposes a Pose encoder and Joint encoder respectively. The Pose encoder predicts multiple whole poses while the joint encoder regards these dispersive poses (kinematic joint points for each posture) as reference points, outputting refined poses as the final results. Due to its progressive learning method, it can refine poses' position output and reach a higher speed and better performance.

Token Pose [16] is another method that uses the Transformer architecture for human posture recognition. Its core idea is to regard the human posture recognition problem as a sequence modeling problem. Specifically, the input image is first converted into a series of "tokens", each token representing a local area of the image. After these tokens are processed by the Transformer, a set of

vectors representing the positions of posture key points are output. The heatmap method is used to predict the positions of posture key points, and each pixel value on the heatmap represents the probability that the position is a posture key point. TokenPose is able to combine key point tokens with their associated patches and key target points contained in the patches. Experiments show that compared with the most advanced CNN-based methods, TokenPose has the same capabilities as similar models and lower weights.

3. Experiment

3.1. Dataset

Datasets serve as the foundation for constructing deep learning. In the task of 2D human pose estimation, the currently commonly used datasets mainly include Buffy, LSP, FLIC, Parse, MPII Human pose, Poses in the wild, MSCOCO, AI Challenger, and Pose Track, and their more detailed information can be found in Table 1.

Table 1. The detail of commonly used 2D Estimation Datasets Concerning Human Pose

Dataset	Size	Type	Num	Description
Buffy	472 frames for training and 276 frames for testing.	Upper body	6 body parts	The data is meticulously obtained from a specific TV program. In addition to that, line segments are thoughtfully provided to clearly indicate the positions of different body parts. What's more, not only are the scale and orientation of the body parts offered, but also some additional details such as the degree of body flexion and extension. It is truly remarkable that in each image, only a single person is annotated, which allows for a focused analysis of individual poses and movements without the distraction of multiple subjects. This unique characteristic makes this dataset particularly valuable for research and applications that require a detailed study of individual human poses.
LSP	There are 1000 images for training and 1000 images for testing.	Full body	14 keypoints	The data originates from Flickr with a sport category tag, and the images are scaled. Similar to the Buffy Dataset, only one person is marked in each image.
FLIC	3987 training images and 1016 testing images.	Upper body	10 keypoints	The data comes from movies, and the people who are occluded or have a non-frontal view are excluded.
Parse	100 images for training and 205 images for testing.	Full body	14 keypoints	The extended annotations not only cover facial expression, gaze direction, and gender but also provide additional details such as age group, emotional intensity, and the direction of head tilt. These comprehensive annotations offer a more in-depth understanding of the subjects in the dataset and can be extremely useful for various research and application purposes related to human behavior analysis and computer vision tasks.
MPII Human pose	Covering 410 activities and 25,000 images.	Full body	16 keypoints	The data is sourced from YouTube videos, which cover 410 human activities, with an activity label to each imagine.
Poses in the wild	30 sequences and 900 frames.	Upper body	5 keypoints	The data is generated from 3 Hollywood movies in the form of 30 video sequences.

MS COCO	115000, 5000, 20000 images for training, validation and test	Full body	17 keypoints	The data is acquired from the Internet and contains a wide variety of activities.
AI Challenger	210000, 30000 and 60000 images for training, validation and test	Full body	14 keypoints	The data is painstakingly crawled from the vast expanse of the Internet. As of now, it stands as the most extensive human pose image dataset available, boasting an extensive collection that encompasses a diverse range of poses, scenarios, and individuals, thereby making it a priceless resource for scholars and practitioners in the field of human pose estimation.
PoseTrack	Comprising 514 videos with a total of 66,374 frames, including 300 videos for training, 50 videos for validation, and 208 videos for testing.	Full body	15 keypoints	The videos are derived from the MPII Human Pose dataset and primarily concentrate on three key aspects: firstly, single-frame multi-person pose estimation, which aims to accurately determine the poses of multiple individuals within a single frame; secondly, multi-person pose estimation in videos, focusing on tracking and estimating the poses of multiple people throughout the video sequence; and finally, multi-person articulated tracking, which involves precisely following the movements of multiple individuals' joints over time.

3.2. Evaluation Metrics

Metrics are invaluable tools that allow us to provide an unbiased assessment of the effectiveness of our refined models. They are not intended to modify model parameters, but rather to facilitate scholars in analyzing various models and determining the most appropriate approach for their distinct applications. As shown in Table 2, a battery of metrics such as Accuracy Percentage, Area Under the Curve, Overall, Kappa Score, Precision Cost Penalty, Precision Cost Kernel, and Precision Discount Jaccard are employed to evaluate these performances.

Part Detection Accuracy reflects robustness, indicative of exceptional model performance. An elevated Pensioner's Category signifies precise joint prediction within set thresholds, demonstrating superior model efficacy. The Area Under the Curve (AUC), a PCK function, objectively evaluates different algorithms. Percentage of Selected Joints (PDJ) corrects shorter limb problems by adjusting the error threshold. The Intersection Over Union (IoU) measures the correlation between predicted and actual bounding boxes using this equation:

$$IoU = \frac{\text{Area of Intersection}}{\text{Area of Union}} \quad (1)$$

Object Keypoint Similarity precisely measures the accuracy of joint estimation through Euclidean distance between predicted and actual joints.

$$OKS = \frac{\sum_i \exp\left(-\frac{2d_i^2}{2s^2 2l_k^2}\right) \delta(v_i > 0)}{\sum_i \delta(v_i > 0)} \quad (2)$$

Table 2. Common metrics of 2D human pose estimation

Metric	Variation	Target	Threshold
PCP	PCP@0.5	Limbs	Reliability of limb measurement multiplied by 0.5
PCK	PCKh	Joints	Reduction in joint clamping or head length by 0.5
AUC	/	Joints	Multiple PCK thresholds utilized
PDJ	PDJ@0.2	Joints	Torso diameter trimmed by 0.2
IOU	/	Joints	Closer approximation of joint clamping box to ground joint
OKS	OKS@0.5, OKS@0.95	Joints	Three OKS thresholds were deployed, including the primary metric (OKS = 0.5:0.05:0.95), the relaxed metric (OKS = 0.5), and the strict metric (OKS = 0.75).
AP	mAP,AP50, AP75, APM,AP L	Joints	Reliability of limb measurement multiplied by 0.5

3.3. Performance Analysis

To further analyze the characteristics of different categories of methods, Table 3 briefly presents the performance of various multi-person pose estimation techniques on the COCO dataset. As shown in Table 3, the bottom-up technique emerges as the most efficient on COCO. The occlusion annotations provided in images from the COCO dataset may contribute to this result [17]. The bottom-up detection algorithm can accurately determine poses in less crowded scenes in the COCO dataset. The results show that among bottom-up methods, SWAIIR's AP value is 72.0, significantly outperforming other algorithms, indicating its optimal performance. In addition, comparing backbone networks, it is evident that many algorithms use IIRNet as a backbone network and exhibit superior performance.

Single-stage methods have their own characteristics in speed and accuracy. As can be seen from the chart, MPE-YOLO has the fastest detection speed and the highest accuracy rate [18]. However, it should be noted that the performance of different algorithms will vary in different scenarios. In practical applications, the appropriate algorithm should be selected according to specific needs. For example, for scenarios with high real-time requirements and not particularly strict requirements for accuracy, MPE-YOLO or OpenPose in the bottom-up method may be a better choice; for scenarios with high requirements for accuracy and less emphasis on speed, Mask R-CNN in the top-down method may be more suitable. In addition, it is also possible to consider combining the advantages of multiple algorithms for mixed use to achieve better results. At the same time, as technology continues to advance further, new algorithms and improvements are constantly emerging, and continuous attention and research are needed to adapt to different application needs and challenges.

Table 3. The comparison of algorithms based on COCO dataset

Method	Backbone	Input size	AP	AP50	AP75	APM	APL
Bottom-up							
Openpose[19]	VGG-19	368	61.8	84.9	67.5	57.1	68.2
PAF[20]	VGG-19	368	60.5	83.4	66.4	55.1	68.1
PersonLab[21]	ResNet152	1401	66.5	88.0	72.6	62.4	72.3
MutiPoseNet[22]	ResNet101	480	69.6	86.3	76.6	65.0	76.3
PifPaf [23]	ResNet152	401	66.7	-	-	62.4	72.3
HigherHRNet[24]	IIRNet-W48	640	68.4	88.2	75.1	64.4	74.2
HigherHRNet[24]	IIRNet-W32	512	66.4	87.5	72.8	61.2	74.2
SWAIIR[25]	IIRNet-W48	640	72.0	90.7	78.8	67.8	77.7
Top-down							
Mask R-CNN[26]	ResNet	800×800	63.1	87.3	68.7	57.8	71.4
LiteHRNet[27]	HRNet-30	384×288	69.7	88.0	75.0	64.3	73.3
G-RMI[26]	ResNet	384×288	68.5	87.1	75.5	65.8	73.3
RMPE[26]	ResNet-18	320×256	61.8	83.7	79.1	68.0	78.6
Single-stage							
PRTR[28]	ResNet-50	384×288	68.2	88.2	75.2	63.2	76.2
PRTR[28]	HRNet-W32	512×384	73.3	89.2	79.9	69.0	80.9
Transformer-based							
TokenPose-Base[29]	Transformer	256×192	74.0	91.9	81.5	70.6	79.8
TokenPose-Large/D24[29]	Transformer	384×288	75.9	92.3	83.4	72.2	80.8
TokenPose-S[29]	Transformer	256×192	73.1	91.4	80.7	69.7	79.0

4. Discussion

Human pose estimation has received widespread attention because it is regarded as a highly practical task in the visual computer field. Though significant performance has been achieved, there are still some challenges:

(1) Limited dataset size. Although existing datasets (such as COCO, MPII, etc.) are large in scale, they still have limitations compared to complex and diverse real-world scenes. At the same time, some postures or scenes appear less frequently in the dataset, resulting in poor performance of the model in these cases. In the process of expanding the dataset size, high-quality labeled data requires professionals to manually annotate the position of each key point, which is time-consuming and laborious. There may be inconsistent annotations between different annotators, resulting in uneven data quality. To address the above problems, the diversity of the dataset can be increased by applying data augmentation technique (such as random cropping, rotation, scaling, etc.). In addition, using a large amount of unlabeled data to assist model training or combining data from multiple modalities such as images, videos, and depth is a feasible solution.

(2) Occlusion and truncation. When the human body is occluded by other objects or its own limbs, it is difficult for the model to detect the key points accurately. When the human body appears at the edge of the image, some limbs may be truncated, resulting in the model being unable to fully recognize all key points. Future work can use the attention mechanism to improve the model's sensitivity to key areas, thereby improving occlusion and truncation problems

(3) Changes in complex backgrounds. In dynamic backgrounds (such as sports fields, streets, etc.), the movement of background objects may interfere with the model's detection. For example, in the case of fast movement, the human body posture changes rapidly, and motion blur or blurred images will make key point detection difficult. In addition, different lighting conditions may impact the image's quality, thereby affecting the accuracy of posture estimation.

5. Conclusion

This paper reviews the human pose estimation task based on deep learning, focusing on the latest research progress of single-person and multi-person human pose estimation. Specifically, based on the various design frameworks of the methods, this paper introduces representative multi-person human pose estimation methods from four aspects: Top-down, Bottom-up, Single-stage and Transformer-based, including their design ideas and key processes. Through comparative analysis on common datasets, this paper analyzes the characteristics of different categories of methods, and summarizes and prospects the current problems and potential directions for future advancement in the field of human pose estimation.

Authors Contribution

All the authors contributed equally and their names were listed in alphabetical order.

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